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Module 5

1) Configuring a Basic Spring Application

pom.xml:

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="
    http://maven.apache.org/POM/4.0.0
    http://maven.apache.org/xsd/maven-4.0.0.xsd">
  <modelVersion>4.0.0</modelVersion>
  <groupId>com.library</groupId>
  <artifactId>LibraryManagement</artifactId>
  <version>1.0</version>
  <dependencies>
    <!-- Spring Core -->
    <dependency>
      <groupId>org.springframework</groupId>
      <artifactId>spring-context</artifactId>
      <version>6.1.8</version>
    </dependency>
  </dependencies>
</project>
```

Repository class:

```
package com.library.repository;

public class BookRepository {

    public void displayBook() {

        System.out.println("Book data fetched from repository");

    }

}
```

Service class:

```
package com.library.service;

import com.library.repository.BookRepository;

public class BookService {

    private BookRepository bookRepository;

    public void setBookRepository(BookRepository bookRepository) {

        this.bookRepository = bookRepository;

    }

    public void showBook() {

        System.out.println("Book Service Called");

        bookRepository.displayBook();

    }

}
```

applicationContext.xml

```
<?xml version="1.0" encoding="UTF-8"?>

<beans xmlns="http://www.springframework.org/schema/beans"

    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
```

```

    xsi:schemaLocation="
        http://www.springframework.org/schema/beans
        https://www.springframework.org/schema/beans/spring-beans.xsd">
<!-- Repository Bean -->
<bean id="bookRepository"
    class="com.library.repository.BookRepository"/>
<!-- Service Bean -->
<bean id="bookService"
    class="com.library.service.BookService">
    <property name="bookRepository"
        ref="bookRepository"/>
</bean>
</beans>

```

Main Class:

```

package com.library;

import com.library.service.BookService;
import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;

public class MainApp {

    public static void main(String[] args) {

        ApplicationContext context =
            new ClassPathXmlApplicationContext(
                "applicationContext.xml");

        BookService service =
            (BookService) context.getBean("bookService");

        service.showBook();

    }
}

```

```
}
```

2) Implementing Dependency Injection

BookRepository.java

```
package com.library.repository;

public class BookRepository {

    public void displayBook() {

        System.out.println("Fetching book details from repository...");

    }

}
```

BookService.java

```
package com.library.service;

import com.library.repository.BookRepository;

public class BookService {

    private BookRepository bookRepository;

    // Setter Injection

    public void setBookRepository(BookRepository bookRepository) {

        this.bookRepository = bookRepository;

    }

    public void showBook() {

        System.out.println("BookService is executing...");

        bookRepository.displayBook();

    }

}
```

applicationContext.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
       xsi:schemaLocation="
           http://www.springframework.org/schema/beans
           https://www.springframework.org/schema/beans/spring-beans.xsd">
    <!-- Repository Bean -->
    <bean id="bookRepository"
          class="com.library.repository.BookRepository"/>
    <!-- Service Bean -->
    <bean id="bookService"
          class="com.library.service.BookService">
        <property name="bookRepository"
                  ref="bookRepository"/>
    </bean>
</beans>
```

Main Class:

```
package com.library;

import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;
import com.library.service.BookService;

public class LibraryManagementApplication {

    public static void main(String[] args) {

        ApplicationContext context =

            new ClassPathXmlApplicationContext(
```

```

        "applicationContext.xml");

    BookService service =

        (BookService) context.getBean("bookService");

    service.showBook();

}

}

```

3) Creating and Configuring a Maven Project

pom.xml

```

<project xmlns="http://maven.apache.org/POM/4.0.0"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="
        http://maven.apache.org/POM/4.0.0
        http://maven.apache.org/xsd/maven-4.0.0.xsd">
    <modelVersion>4.0.0</modelVersion>
    <groupId>com.library</groupId>
    <artifactId>LibraryManagement</artifactId>
    <version>1.0-SNAPSHOT</version>
    <dependencies>
        <!-- Spring Context -->
        <dependency>
            <groupId>org.springframework</groupId>
            <artifactId>spring-context</artifactId>
            <version>5.3.30</version>
        </dependency>
        <!-- Spring AOP -->
        <dependency>

```

```
<groupId>org.springframework</groupId>
<artifactId>spring-aop</artifactId>
<version>5.3.30</version>
</dependency>
<!-- Spring Web MVC -->
<dependency>
  <groupId>org.springframework</groupId>
  <artifactId>spring-webmvc</artifactId>
  <version>5.3.30</version>
</dependency>
</dependencies>
<build>
  <plugins>
    <plugin>
      <groupId>org.apache.maven.plugins</groupId>
      <artifactId>maven-compiler-plugin</artifactId>

      <version>3.11.0</version>
      <configuration>
        <source>1.8</source>
        <target>1.8</target>
      </configuration>
    </plugin>
  </plugins>
</build>
</project>
```